

Components for Optical Communication Systems

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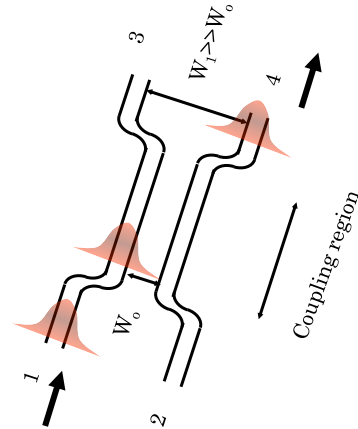
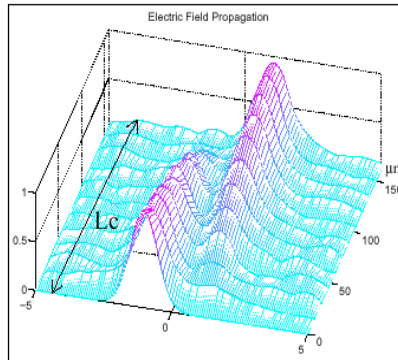
1

Optical Directional Coupler

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2

Theory of Directional Coupler (DC)



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3

Theory of Directional Coupler

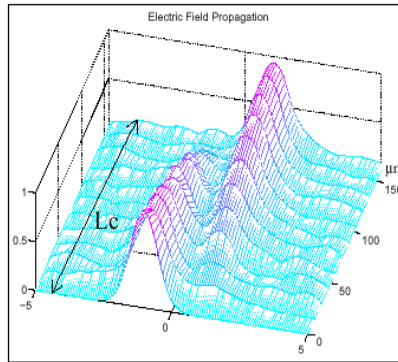
***Due to the open nature of the dielectric optical waveguides, when two guides are brought close to each others, optical field propagating in one of them could be coupled to the other.**

***This coupling phenomenon can be explained by the existence of a tail to the optical field outside the guide core. This tail excites the modes of the adjacent guide and thus results in the power transfer from one guide to the other.**

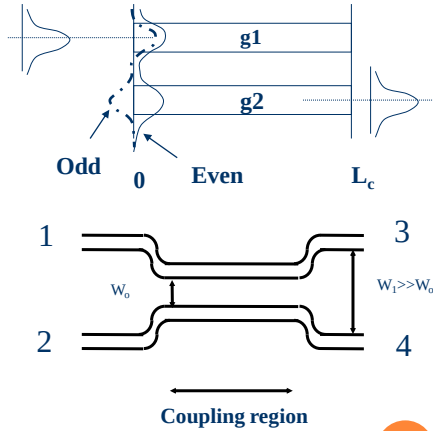
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4

Theory of Directional Coupler



$$L_c = \frac{\pi}{\beta_e - \beta_o}$$



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5

Theory of Directional Coupler

***This coupling can also be explained by the modal interference in the overall guides consisting of the two single mode guides.**

***At the input, we excite one of the guides and thus we excite the two modes (even and odd modes) of the overall guiding structure with equal phase.**

***When the two modes are out of phase, this will be equivalent to the optical power transfer to the second guide. The coupling length is thus the length corresponding to a phase shift of 180 degrees between the two modes.**

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6

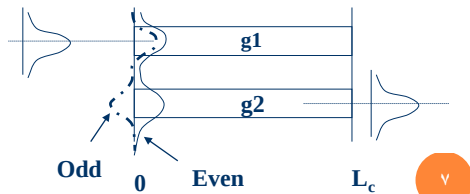
MODE INTERFERENCE

- Electric field in waveguide can be approximated by the summation of the first even and odd modes of the whole structure

$$E(x, z) = E_e(x)e^{-j\beta_e z} + E_o(x)e^{-j\beta_o z}$$

$$|E(x, z)| = |E_e(x) + E_o(x)e^{j(\beta_e - \beta_o)z}|$$

- $|E(x, 0)| = |E_e(x) + E_o(x)| = E_1(x)$
- $|E(x, L_c)| = |E_e(x) - E_o(x)| = E_2(x)$



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7

MODE INTERFERENCE

*The system of the two SM guides g1 and g2 could be considered as a single wide guiding structure having 5 layers. This wide guiding structure will have at least two guided modes, an even fundamental mode and a higher order odd mode.

*These two modes are illustrated in Fig. in the plane $z=0$ at the waveguide input. When one of the individual SM guides (say g1) is excited by its fundamental mode we actually excite the two modes the even mode E_e and the odd mode E_o of the 5-layer guiding structure with nearly equal power and the same phase at $z=0$.

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8

MODE INTERFERENCE

***When the optical field propagates in the structure, the two modes propagate with different velocities according to their propagation constants. The phase difference between the two modes results in a power redistribution in the guiding structure such that at a certain length L they could be out of phase.**

***The sum of the two modes when they are out of phase results in a field localized in the second guide g2 in the plan $z=L$. This length thus corresponds to the coupler coupling length.**

9

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9

MODE INTERFERENCE

○The coupling length

$$L_c = \frac{\pi}{\beta_e - \beta_o}$$

○The coupling coefficient

$$\kappa = \frac{\pi}{2L_c} = \frac{\beta_e - \beta_o}{2}$$

Then $L_c = \pi/2\kappa$

○ β_e (even), β_o (odd) are calculated through solving the 5 layer structure.

***The coupling length can thus be calculated as the length corresponding to π phase shift between the modes and consequently the coupling coefficient could be defined as shown.**

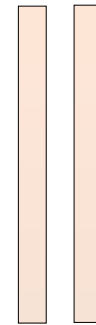
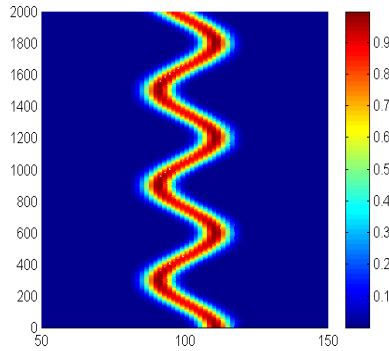
10

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10

PROPAGATION IN DIRECTIONAL COUPLER

- Important: Note that the field is periodic



DC



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11

DIRECTIONAL COUPLER AS A CIRCUIT ELEMENT

To derive an equivalent circuit model for the coupler, the field in guide g1 can be considered as the sum of the two modes while the field in g2 is the difference between them.

Assume the input at port 1 only, for symmetric coupler we have

$$E_I = E_e e^{-j\beta_e z} + E_o e^{-j\beta_o z} \quad \text{where} \quad E_e J_{\text{peak}} = E_o J_{\text{peak}} = E_{\text{in}}/2$$

This can be written as

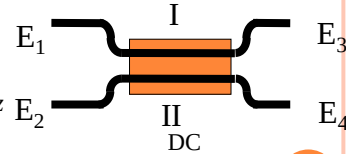
$$E_I = \frac{E_{\text{in}}}{2} e^{-j(\beta_e + \beta_o)z/2} \left[e^{-j(\beta_e - \beta_o)z/2} + e^{+j(\beta_e - \beta_o)z/2} \right]$$

Or simply:

$$E_I = \frac{E_{\text{in}}}{2} e^{-j\phi_{\text{av}}} \left[e^{-j\Delta\phi} + e^{+j\Delta\phi} \right]$$

with

$$\phi_{\text{av}} = \frac{\beta_e + \beta_o}{2} z \quad \text{and} \quad \Delta\phi = \kappa z = \frac{\pi}{2L_c} z = \frac{\beta_e - \beta_o}{2} z$$



Thus we get $E_I = 2E_e \cos(\Delta\phi)$

i.e. $E_I = a E_1$ with $a = \cos(\Delta\phi) = \cos(\pi z/2L_c)$



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12

DIRECTIONAL COUPLER AS A CIRCUIT ELEMENT

- Similarly in Guide II we have

$$E_{II} = E_e e^{-j\beta_e z} - E_o e^{-j\beta_o z}$$

$$E_{II} = E_e e^{-j\phi_{av}} (e^{-j\Delta\phi} - e^{+j\Delta\phi}) = -j2E_e \sin(\Delta\phi)$$

- Thus, in General, for the input at I we have:

$$E_3 = a E_1 \quad \text{and} \quad E_4 = -jb E_1$$

with $a = \cos(\pi L/2L_c)$, $b = \sin(\pi L/2L_c)$

- Using the superposition

For an arbitrary input excitations E_1 and E_2 at ports 1 and 2 respectively, the output fields at ports 3 and 4 can be written as:



$$E_3 = aE_1 - jbE_2$$

$$E_4 = -jbE_1 + aE_2$$

- For loss-less coupler $a^2 + b^2 = 1$

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13

DIRECTIONAL COUPLER AS A CIRCUIT ELEMENT

*The optical directional coupler can be considered as a 4-port element that can be used as an optical beam splitter/combiner.

*An input optical signal at port 1 (or 2) will be coupled to ports 3 and 4 only according to its coupling coefficient, where a^2 is the power forward coupling coefficient between the connected arms, b^2 is the power cross coupling coefficient between the coupled arms and the common phase shift between the two outputs E_3 and E_4 is omitted.

*For a 3 dB coupler, the optical power injected in port 1 is divided equally between ports 3 and 4 and the power coupled to port 2 is negligible (actually it represents the directivity or the isolation of the coupler). Similarly, a power injected at port 3 (or 4) will be divided equally between ports 1 and 2

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14

DC Properties

- Note that:
 - The two outputs of the DC have a $\pi/2$ phase shift
 - We can control the ratio of the outputs by controlling the coupler length
 - The coupling coefficient depends on the wavelength, thus the coupler can be used as a DeMux.

$$L_c = \frac{\pi}{\beta_e - \beta_o} = \frac{\lambda}{2(n_{ee} - n_{eo})}$$

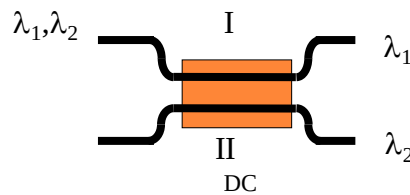
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15

DC Mux/ DeMux (WDM coupler)

- Design the DC such that
 - $L = nL_c$ for λ_1 & $L = mL_c$ for λ_2
with n even, m odd or n odd, m even

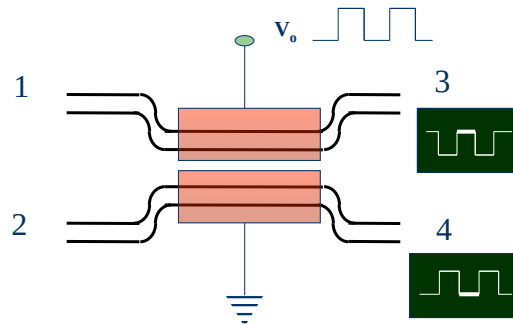


١٦

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16

DC Electro-Optic Modulator



V_0 changes $\Delta\beta \rightarrow$ coupling between guides

Applied electric field modifies the refractive index $\rightarrow$ It modifies relative velocities *and* then the coupling between waveguides such that it switches the output between port 3 and port 4. This could be useful in different applications such as **Modulators, Routers, Switch, ...etc**

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17

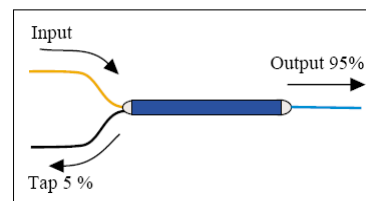
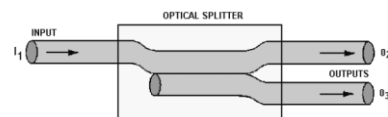
FIBER DIRECTIONAL COUPLER

Directional Coupler types

- 2x2 directional coupler
- Splitter 1x2
- Tap Coupler
- WDM couplers

Function of tap coupler: Taping a low fraction (5-10 %) of optical power.

Main Application: Monitoring



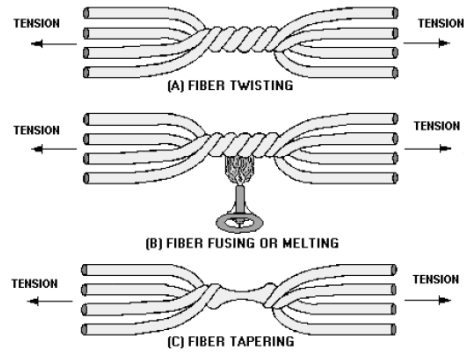
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18

Coupler Fabrication

○ Fused Coupler fabrication:

- Twisting
- Melting
- Tapering



19

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